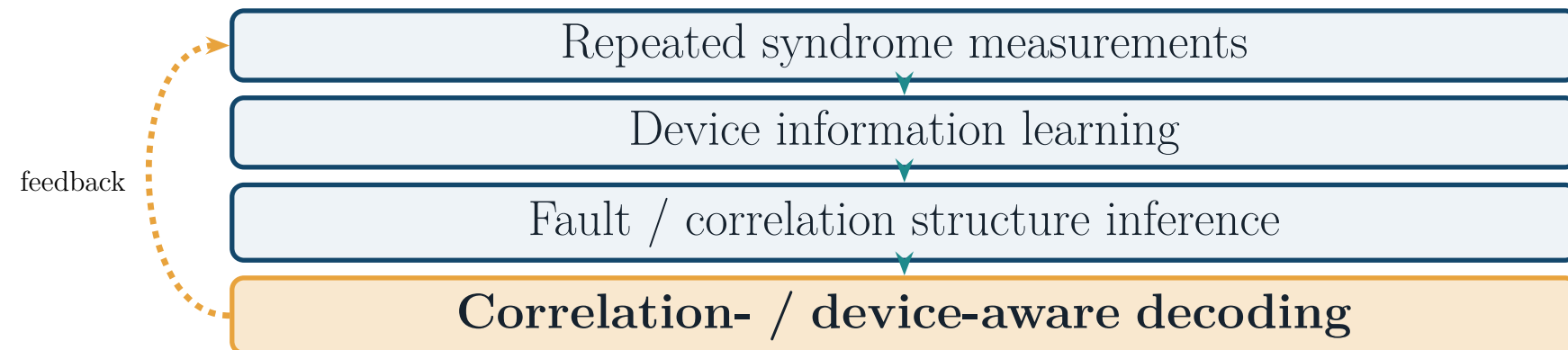


1. Motivation

Surface codes already measure syndromes every round—bits generated for decoding, then discarded.

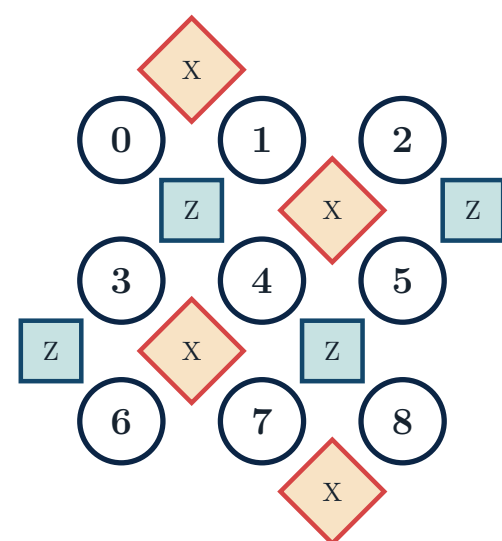
“Can we read the device’s own fault structure out of data we already collect for QEC?”

Long-term vision: a **closed loop** recycling QEC data into device knowledge, then back into better decoding:



2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits); each extraction round issues **24 directed CNOTs**, one of which is the *dominant* fault source.



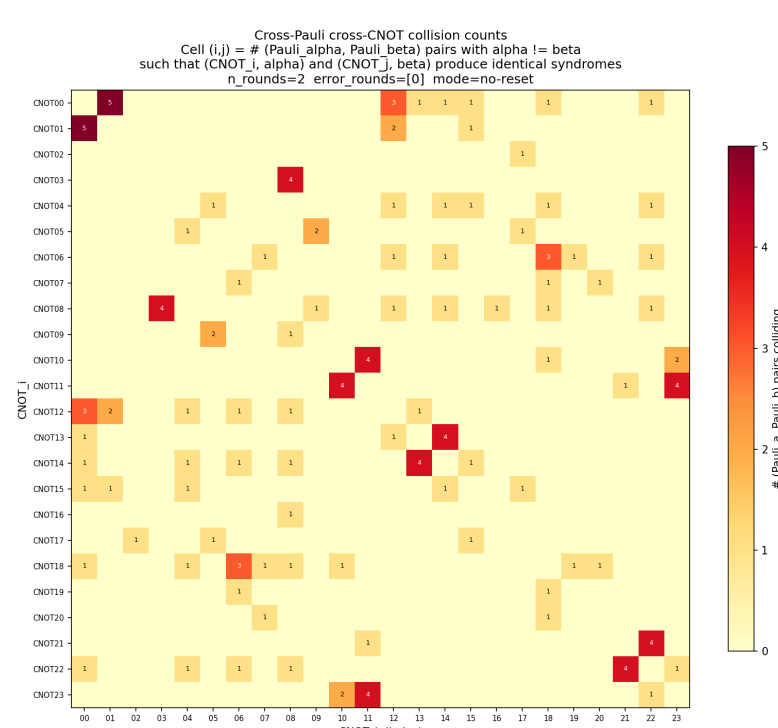
Rotated $d=3$ layout: 9 data qubits (circles), 8 Z-/X-stabilizer ancillas (squares / diamonds).

The question

From the syndrome sequence $S = (s_1, \dots, s_T)$ alone, identify the **dominant faulty CNOT** among the 24—with the fault’s rounds and Pauli type (XX, ..., ZZ) unobserved.

3. Single-Shot Ambiguity

Is one measurement enough? **No.** With the Pauli type unknown, distinct (CNOT, Pauli) faults produce *identical* syndromes:



Collision map: cell (i, j) counts Pauli pairs for which faults at $CNOT_i$ and $CNOT_j$ are indistinguishable.

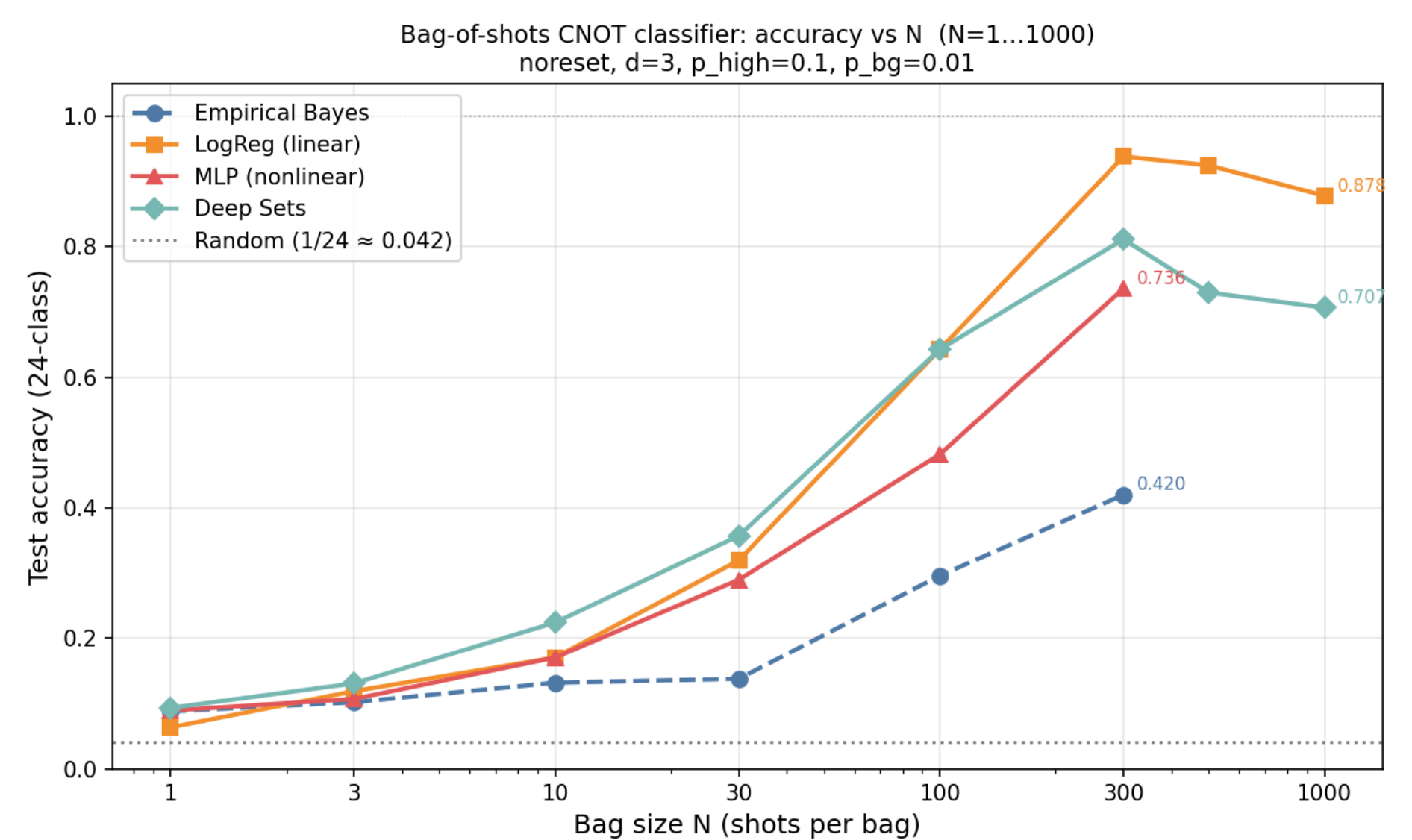
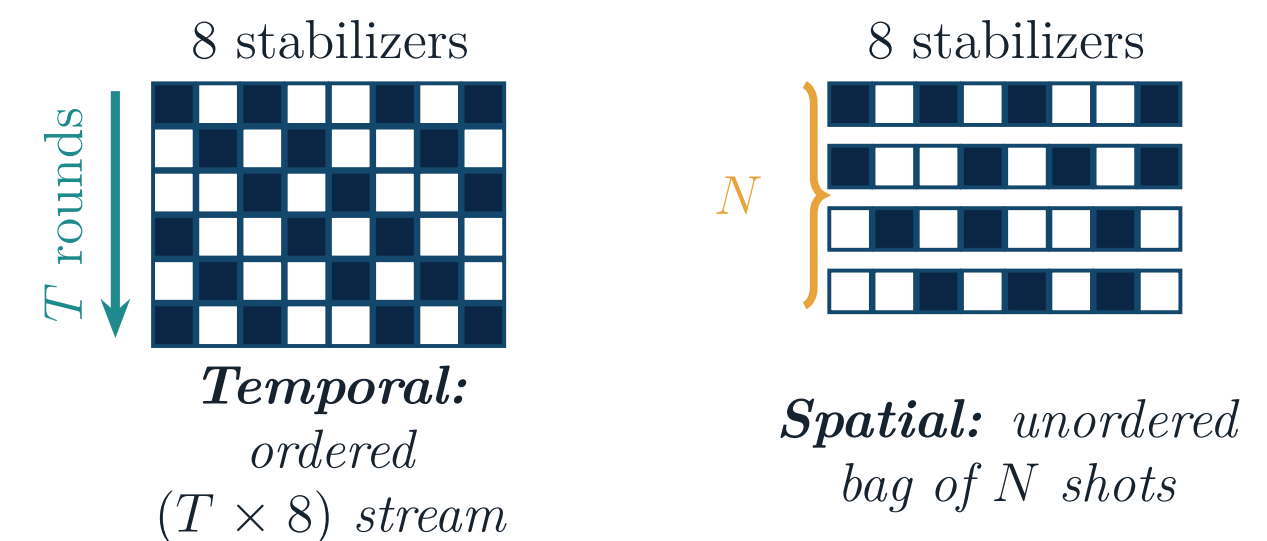
216 → **123** **72**
unique signatures cross-CNOT collision pairs

Single-shot 24-class CNOT identification is information-theoretically impossible.

Invariant under round count and reset mode.

4. Temporal Information Breaks the Ambiguity

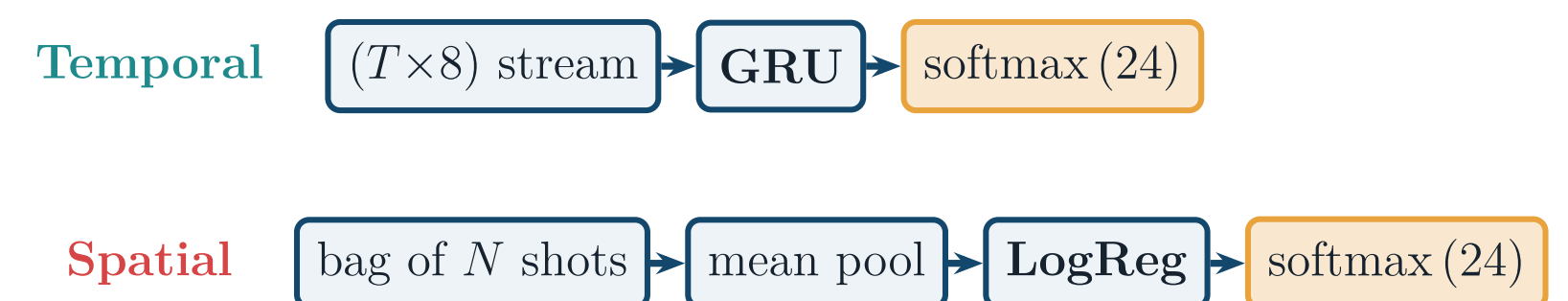
The way out is **temporal structure**: reading the fault across many rounds. The same fault is probed either *temporally* (a long syndrome stream) or *spatially* (a bag of repeated single-round shots):



Accuracy vs bag size N . At $N=1$ (single shot) accuracy is near random ($1/24$); by $N=300$ it reaches ≈ 0.9 —identifiability is restored by **accumulating information across rounds/shots**.

5. Learned Decoders

Two matched model classes, one per regime:



Best accuracy preliminary · $d=3$

$\approx 95\%$ Temporal GRU **$\approx 94\%$** Spatial LogReg
($N=300$ shots/bag)

The GRU breaks the rule-based single-round ceiling (0.75).

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at $d=3$.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop—feed inferred fault / correlation structure into **device-aware decoding**.